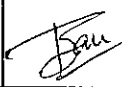
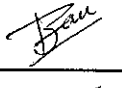
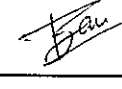
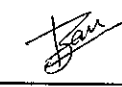
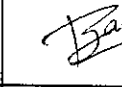


PTVA'S
MULUND COLLEGE OF COMMERCE (AUTONOMOUS)
DEPARTMENT OF INFORMATION TECHNOLOGY
T.Y.B.Sc. Information Technology

Name: Sawant Aarush Hemant

Roll No: 269729

Course Name: INTERNET OF THINGS Practical

Sr. No.	PRACTICAL TITLE	DATE	SIGNATURE (Attendance)	SIGNATURE (Submission)
0.	Starting Raspbian OS, Familiarizing with Raspberry Pi Components and interface, Connecting to ethernet, Monitor, USB.	24/06/2026		
1.	Displaying different LED patterns with Raspberry Pi.	30/06/2026		
2.	Visitor Monitoring with Raspberry Pi and Pi Camera	21/07/2026		
3.	Controlling Raspberry Pi with Telegram bot.	30/07/2026		
4.	Installing Windows 10 IoT Core on Raspberry Pi	14/07/2026		
5.	Raspberry Pi GPS Module Interfacing			
6.	Displaying Time over 4-Digit 7-Segment Display using Raspberry Pi.			
7.	IoT based Web Controlled Home Automation using Raspberry Pi			
8.	Raspberry Pi Based Oscilloscope			
9.	Interfacing Raspberry Pi with RFID			
10.	Fingerprint Sensor interfacing with Raspberry Pi			

Final Submission: _____

Sr.No	Date	Topic	Sign
0	24/6/26	Practical 0	<i>[Signature]</i>

Aim : Starting Raspbian OS, Familiarizing with Raspberry Pi Components & interface, Connecting to ethernet, Monitor, USB.

Abstract : The Raspberry Pi is a mini computer specifically created to make tech learning easier. It has a lot of components for computer-based projects, like USB ports, an Ethernet port, an SD card slot, Wi-Fi antenna ports & more.

Required Components :

A) Hardware

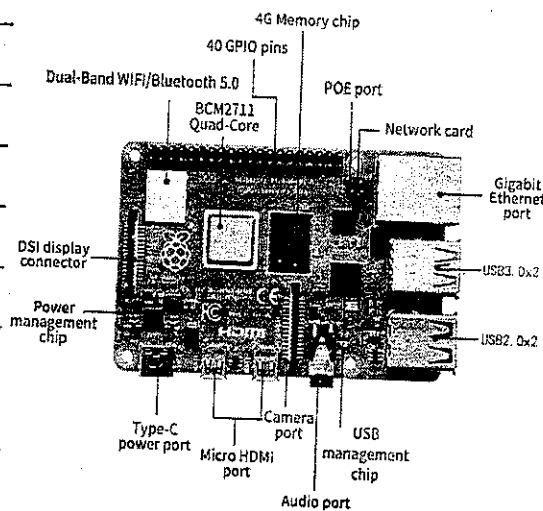
Particular	Quantity
Raspberry Pi	1
Power Supply	1
USB Keyboard	1
USB Mouse	1
Micro SD Card	1
Micro SD USB card reader	1
A monitor that supports HDMI	1
HDMI cable	1
Ethernet cable	1

B) Software

Raspbian :-

Pto. →

Raspberry Pi 2 GPIO Header			
Pin#	NAME		NAME Pin#
01	3.3v DC Power		DC Power 5v 02
03	GPIO02 (SDA1, PC)		DC Power 5v 04
05	GPIO03 (SCL1, PC)		Ground 06
07	GPIO04 (GPIO_GCLK)		(TXD0) GPIO14 08
09	Ground		(RXD0) GPIO15 10
11	GPIO17 (GPIO_GEN0)		(GPIO_GEN1) GPIO18 12
13	GPIO27 (GPIO_GEN2)		Ground 14
15	GPIO22 (GPIO_GEN5)		(GPIO_GEN4) GPIO23 16
17	3.3v DC Power		(GPIO_GEN5) GPIO24 18
19	GPIO10 (SPI_MOSI)		Ground 20
21	GPIO09 (SPI_MISO)		(GPIO_GEN8) GPIO25 22
23	GPIO11 (SPI_CLK)		(SPI_CE0_N) GPIO28 24
25	Ground		(SPI_CE1_N) GPIO37 26
27	ID_SD (PC ID EEPROM)		(PC ID EEPROM) ID_SC 28
29	GPIO05		Ground 30
31	GPIO06		GPIO12 32
33	GPIO13		Ground 34
35	GPIO19		GPIO16 36
37	GPIO26		GPIO20 38
39	Ground		GPIO21 40



1. USB ports :- These are used to connect a mouse & keyboard. You can also connect other components like USB drive.
2. SD Card Slot :- One can slot the SD card here. This is where the OS software & your files are stored.
3. Ethernet port :- This is used to connect the Raspberry Pi to network with a cable. The Raspberry Pi can also connect to a network via wireless LAN.

4. Audio jack :- You can connect headphones or speakers here.
5. HDMI port :- Here monitor/projector is connected that one will use to display output from the Raspberry Pi. If your monitor has speakers you can also use them to hear sound.
6. Micro USB power connector :- Here one connects a power supply. You should always do this last, after connecting all other components.
7. GPIO ports :- These allow you to connect electronic components like LEDs & buttons to Raspberry Pi.

C) Procedure

1. Insert SD card into card reader.
2. Open file explorer. Go to 'D' or 'E' drive. Open 'Softwares' folder.
3. In softwares, right click on rufus.exe file & click 'open'. User control access is asked. Click on 'yes'. Automatically full window will open.
4. Check and name in device. If displayed, do as under:
 - a. In Boot selection, click on 'select'. Then browse raspberry pi compressed folder as "2021-03-15-raspbian-bookworm-amhf-full.img.xz" folder.
 - b. After folder selection, click 'Start'. Check status of file transfer. It shows 100% in status.

c. Click 'close'.

5. Insert Card in Raspberry Pi.

6. Connect mouse, keyboards to USB ports of Raspberry Pi.

7. Connect Raspberry Pi to HDMI port directly or use HDMI to VGA connector.

8. Connect Pi to internet via Ethernet, use an Ethernet cable to connect the Raspberry Pi.

9. Plug power supply into socket & connect it to micro-USB power port.

10. Red light means power On. Green means the booting of Pi. It'll boot up into a desktop.

11. Set up of the Raspberry Pi.

12. Configuring your Pi:

a. You can control most of your Raspberry Pi's settings (password) through its config. app. found in preference or menu.

b. System: In this tab, you change basic sys. settings of Pi.

i) Boot: Select to show Desktop/CLI when Raspberry Pi starts.

ii) Auto Login: Enabling this will make Pi login automatically whenever started.

iii) Network at Boot: This will cause your Pi to wait until a network connection is available before starting.

iv) Splash Screen: Choose whether to show splash (startup) screen when your Pi boots.

v) Resolution: You can set screen resolution here.

vi) Underscan: Choose whether your Pi should show black bars at top & bottom of screen when it can't match screen resolution.

c. Interfaces: You can link devices & components to Raspberry Pi using lots of diff. types of connections. The interface tab is where you turn these diff. connections on/off, so that Pi recognizes that you've linked something to it via a particular type of connection.

i) Camera: Enable Raspberry Pi Camera Module.

ii) SSH: Allow remote access to Pi from another computer using SSH.

iii) VNC: Allow remote access to Pi Desktop from another computer using VNC.

iv) SPI: Enable SPI GPIO pins.

v) I2C: Enable I2C GPIO pins.

vi) Serial: Enable the serial (Rx, Tx) GPIO pins.

vii) Remote GPIO: Allow access your Raspberry Pi's GPIO pins from another computer.

d. Performance: If needed to do for a project, change performance settings of your Pi this tab.

i) Overclock: Change CPU speed & voltage to increase performance.

ii) GPU Memory: Change allocation of memory given to GPU.

This tab allows you to change your Pi settings to be specific to a country/location.

Locale: Set language, country & character set used by your Raspberry Pi.

Time zone: Set the time zone.

Keyboard: Change the keyboard layout.

Wi-Fi Country: Set the Wi-Fi country code.

13. After starting with Raspberry Pi for the first time, 'Welcome to Raspberry Pi application' will pop up & it will guide through initial setup.

14. Click 'Next' to start setup.

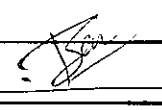
15. Set respective Country, Language & Time Zone, then again 'Next'.

16. Enter a new password & 'next'.

17. Connect to your Wi-Fi.

18. Click next & don't update.

19. Click Done/Reboot to finish setup.

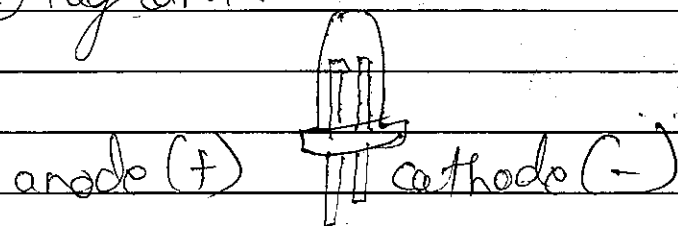
Sr.No	Date	Topic	Signature
1	30/6/26	Displaying different LED patterns with Raspberry Pi	

Aim: Displaying different LED patterns with Raspberry Pi.

Hardware needs:-

Raspberry Pi, Ethernet, Jumper wires (Female to Female) & LED

Diagram:-



Code:-

```
import RPi.GPIO as GPIO
import time
```

```
# Get user input
```

```
numTimes = int(input("Enter total no. of times to blink:"))
speed = float(input("Enter length of each blink (sec):"))
```

```
GPIO.setwarnings(False)
```

```
GPIO.setmode(GPIO.BOARD) # Using physical pin
# nos
```

```
# Setup GPIO pins for output
```

```
GPIO.setup(11, GPIO.OUT) # LED 1
```

```
GPIO.setup(13, GPIO.OUT) # LED2
```

```
def Blink(numTimes, speed):  
    for i in range(numTimes):  
        print("Iteration", i)
```

```
        GPIO.output(11, True) # LED1 ON  
        GPIO.output(13, True) # LED2 ON  
        time.sleep(speed)
```

```
        GPIO.output(11, False) # LED1 OFF  
        GPIO.output(13, False) # LED2 OFF  
        time.sleep(speed)
```

```
Blink(numTimes, speed)
```

```
GPIO.cleanup()  
print("Done")
```

Sr.No.	Date	Topic	Sign
2	21/7/26	Visitor Monitoring with Raspberry Pi + Pi Camera	

Commands

1) `sudo raspi-config`

Purpose: Opens Raspberry Pi Configuration Menu

Explanation: `sudo` gives administrator permission & `raspi-config` is used to configure Pi's settings. Used to enable/configure camera related settings, interfaces, sys-options, etc.

2) `sudo reboot`

Purpose: Restarts Pi

Explanation: After making config changes, Pi needs to be restarted for changes to take effect.

3) `libcamera -h`

Purpose: Checks if Pi camera functions.

Explanation: Starts camera & shows a preview window. Used to check camera connection & op.

4) `libcamera -still -o image.jpg`

Purpose: Captures a still img. using Pi Camera.

Explanation: `libcamera -still` is used to take photographs. `-o` specifies output file name. The captured img. is saved as `image.jpg`.

Output: An img. file named `image.jpg` is created.

- 5) `libcamora -vid -o video 426h -t 10000`.
Purpose: Records a video using Pi Camera.
Explanation: `libcamora -vid` is used for video recording. `-o` specifies output file, while `-t 10000` specifies recording time in milliseconds.
Output: A video file named `video 426h` is created.



Sr.No.	Date	Topic	Sign
3	30/7/26	Controlling Raspberry Pi with Telegram	

1) Open Telegram in your mobile.
Start "BotFather".

2) Open "BotFather".

3) /start

4) Create a new Bot

5) Obtain access token

6) Install "Python Package Index"
sudo apt-get install python-pip

7) Install "telepot"
(To download telepot, create a virtual environment).

1. Installing the venv module:-

sudo apt install python3-venv

2. Creating a new virtual environment. Myenv is user-defined name

python3 -m venv myenv

3. Activate the virtual environment

source myenv/bin/activate

4. It should look like :- (myenv)pi@raspberrypi:~\$

5. Install telepot in VM:-

pip install telepot
6. pip install RPi.GPIO

- 8)
1. Clone the git
git clone <https://github.com/salmanfarisvp/TelegramBot.git>
 2. Go into the cloned repository
cd TelegramBot
 3. Edit the code to add your bot token
nano telegrambot.py
 4. After this the .py file will open. Paste bot token here
bot = telepot.Bot('your_bot_token')
 5. Run the code
python telegrambot.py

a) Connect LED to Pi

10) Send Command

Start our Bot

Send "on" & "off"

Look at your Pi, you can see the LED on & off when you send "on" & "off" to our bot.

Code:-

(This code will be displayed after the command nano telegrambot.py)

```
import sys
import time
import random
import datetime
import telegram
import RPi.GPIO as GPIO
```

```
# LED
```

```
def on(pin):
```

```
    GPIO.output(pin, GPIO.HIGH)
    return
```

```
def off(pin):
```

```
    GPIO.output(pin, GPIO.LOW)
    return
```

```
# to use Raspberry Pi board pin numbers
```

```
GPIO.setmode(GPIO.BOARD)
```

```
# set up GPIO output channel
```

```
GPIO.setup(11, GPIO.OUT)
```

```
def handle(msg):
```

```
    chat_id = msg['chat']['id']
    command = msg['text']
```

```
    print 'Got command: %s' % command
```

```
    if command == 'on':
```

```
        bot.sendMessage(chat_id, on(11))
```

```
    elif command == 'off':
```

```
        bot.sendMessage(chat_id, off(11))
```

```
bot = telepot.Bot('Bot Token')  
bot.message_loop(handle)  
print 'I am listening ...'
```

```
while 1:  
    time.sleep(10)
```

1) To deactivate virtual environment
deactivate

Sr. No	Date	Topic	Sign
4	14/7/28	Installing Windows 10 ToT Core on Raspberry Pi	<i>[Signature]</i>

Aim: To install Windows 10 on Raspberry Pi 4.

Hardware:

Raspberry Pi 4, MicroSD card, Ethernet adapter (optional), HDMI monitor, keyboard, mouse

Step 1) Download latest pre-release version of WoR tool & extract files.

Step 2) Prepare Windows 10 ARM64 ISO

1. Go to UUPDump.net
2. Choose Windows 10 ARM64 build (choose latest - stable)
3. Download script & run it on your Windows PC.
4. At end, you'll have Win 10 ARM64 ISO file.

Step 3) Install WoR & Prepare the SD/SSD.

1. Insert your SD card in Windows PC.
2. Open WoR.exe (run as Admin)
3. Follow wizard:
 - i) Point to Win 10 ARM64 ISO
 - ii) Select device type: Raspberry Pi 4
 - iii) Give drivers & UEFI firmware
 - iv) Confirm installation settings
4. WoR will now flash Win10 img. to your device. This can take upto an hour.

Step 4) Booting Raspberry Pi 4.

1. Insert SD card in Pi.
2. Power it on $\rightarrow$ it'll load into UEFI firmware screen.
3. (Optional but recommended)
 - i] Go into UEFI settings
 - ii] Change Boot Order to make SD first boot option.
 - iii] Save & reboot.
4. Now, Win 10 setup starts.
 - i] Select language, region, keyboard layout
 - ii] Create or skip Microsoft acc.
 - iii] Config. sys preferences.
5. For network access, use USB to Ethernet or Wifi Dongle.
6. Install Microsoft Edge by double clicking on Microsoft Edge desktop icon.

